IGKNIGHT — Project Summary

The Vision

IGKNIGHT is a gamified digital wallet web application that transforms campus recycling at Université Saint-Joseph (USJ) into an engaging, rewarding, and community-driven experience. By merging environmental sustainability with Jesuit values and medieval knight-themed gamification, IGKNIGHT turns every recycled item into a quest, every student into a knight, and every act of service into a step toward a greener future.

The Problem We're Solving

University campuses generate significant amounts of recyclable waste daily — plastic bottles, paper, glass, aluminum cans — yet most of it ends up in general waste bins.
Students lack motivation, awareness, and tangible incentives to recycle consistently.
Traditional recycling programs rely on posters and goodwill alone, which fail to create lasting behavioral change.
At the same time, student clubs and campus organizations constantly struggle to fund their activities and events, relying on small budgets and sporadic donations.
IGKNIGHT solves both problems simultaneously by creating a circular economy where recycling generates digital currency that students can use to support their communities, pay for campus services, and track their collective environmental impact.

How It Works

The Cycle

Student recycles at a Smart Bin on campus
↓
Earns Scudo (⛨) — digital coins in their IGKNIGHT wallet
↓
Spends Scudo in two ways:
1. Buy USJ Merchandise (Shop)
2. Invest in Social club projects

↓
Meanwhile, IGKNIGHT volunteers collect the physical waste
↓
Waste is taken to LWM (Lebanese Waste Management) drive-through
↓
LWM pays real money per kilogram of recyclable waste

↓
That real money is distributed to clubs proportional
to the Scudo pledged by students (or prepare the merchandises)
↓
Clubs fund their events and activities
↓
Students see their impact, earn achievements, level up
↓
The cycle continues — recycling funds community life

This creates a self-sustaining ecosystem where no external funding is needed. The waste itself generates the value. Students are motivated by rewards. Clubs are motivated by funding. The environment benefits from every transaction.

The Smart Bin

IGKNIGHT integrates software with hardware. A physical Smart Bin prototype built with Arduino is equipped with a servo-controlled lid, and LED indicators. Students interact with it through the web app:

1. Select waste type and items in the app
2. Scan the QR code on the bin
3. The app communicates with the Arduino
4. The bin lid opens physically
5. Student deposits recyclable waste
6. Bin closes, and Scudo is automatically credited

This hardware-software loop makes recycling feel tangible, interactive, and rewarding — not like a chore but like completing a mission.

How IGKNIGHT Responds to Sustainability

Environmental Sustainability

- Waste Diversion: Every item recycled through IGKNIGHT is diverted from landfill and channeled to proper recycling through the LWM partnership
- Impact Measurement: The app tracks kilograms recycled and converts them to CO_2 offset using standard environmental conversion factors, making invisible impact visible
- Behavioral Change: Gamification (achievements, leaderboards) creates habits, not one-time actions. Research consistently shows that reward-based systems produce lasting behavioral shifts in recycling participation
- Campus-Wide Accountability: Faculty and department competitions encourage entire academic communities to participate, not just environmentally conscious individuals

Economic Sustainability

- Self-Funding Model: Unlike programs that require continuous university budget allocation, IGKNIGHT generates its own economic value through the LWM

waste-to-cash pipeline. The more students recycle, the more real money flows into the system
- Minimal Operational Cost: The app is built entirely on free-tier infrastructure (Vercel hosting, Supabase database), requiring no financial investment to maintain, just an initial funding to buy the Arduino components at first.
- Club Funding Pipeline: Student organizations gain a new, sustainable funding source tied directly to community participation rather than administrative budgets

Social Sustainability

- Community Building: The pledge system connects individual action to collective benefit — when a student recycles and invests Scudo to a club, they are directly enabling campus community life
- Inclusive Participation: Every student can contribute regardless of financial status — recycling costs nothing but generates value for everyone
- Volunteer Engagement: The bulk collection system creates meaningful volunteer opportunities for students who want to contribute beyond personal recycling, and encourage in home recycling.

How IGKNIGHT Embodies Jesuit and USJ Values

Magis — The Pursuit of More

The Jesuit principle of Magis calls us to always strive for the greater good, to do more, to go further. IGKNIGHT embeds this directly into its design:

- The "Leaderboard" section shows collective campus impact, inspiring students to push beyond individual achievement toward community-wide transformation
- Every quest completed is an act of Magis — choosing to recycle when it would be easier not to, choosing to sort waste when a general bin is closer

Cura Personalis — Care for the Whole Person

Cura Personalis is the Jesuit commitment to caring for the entire person — mind, body, and spirit. IGKNIGHT reflects this by:

- Engaging the mind: Students learn about waste types, environmental impact, and CO_2 conversion through the app's data and feedback
- Engaging action: The physical interaction with smart bins and the tangible reward system connect knowledge to behavior
- Engaging community: The invest systems move students from self-interest to service, developing them as whole persons who act for others

Men and Women for Others

This foundational Jesuit value — forming people who live not for themselves but for the service of others — is the heartbeat of IGKNIGHT:

- The invest system is designed so that personal earnings (Scudo) are most meaningful when given away to support club campaigns

- The volunteer system creates opportunities for students to serve the campus community by collecting and processing bulk recyclables
- Students don't just recycle for personal reward — they recycle to fund their peers' activities, events, and initiatives

USJ's Mission

USJ's commitment to forming responsible citizens who contribute to Lebanese society and the common good is directly served by IGKNIGHT:

- It addresses a real Lebanese environmental challenge (waste management) with a locally designed solution
- It partners with LWM (Lebanese Waste Management), grounding the project in Lebanon's existing infrastructure
- It builds a model that could expand beyond USJ to other universities and communities across Lebanon
- It demonstrates that innovation, technology, and values can work together — a core USJ aspiration

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The Names — Origin and Meaning

IGKNIGHT

The name operates on three layers:

Layer	Reading	Meaning
Historical	IG + KNIGHT	Ignatius the Knight — Before his conversion, St. Ignatius of Loyola was a knight and soldier. The name honors his original identity and the journey that began from it
Transformational	IGNITE	To ignite — The cannonball that struck Ignatius at the Battle of Pamplona in 1521 shattered his leg but ignited his spiritual transformation. IGKNIGHT represents that same spark — a small action (recycling) that ignites massive change (environmental and community impact)
Active	IG + KNIGHT (verb)	To knight someone — The app "knights" students into environmental service, elevating everyday people into agents of change. Every student who joins becomes a knight in a green crusade

The name carries the full arc of the Ignatian story: a knight is struck down, ignited by a new vision, and rises to change the world. IGKNIGHT invites students into that same narrative — be struck by the reality of environmental crisis, be ignited by the possibility of change, and rise as knights who act.

Scudo (⛨)

The digital currency is named Scudo for deeply intentional reasons:

Layer	Meaning
Historical Authenticity	The scudo was a real coin used across Italian states during the 15th-16th centuries — the exact period when Ignatius of Loyola lived (1491-1556). It was the currency of his world, the coin a knight would have carried
Literal Translation	Scudo means "shield" in Italian. A shield is the knight's primary tool of protection — just as recycling is our shield protecting the environment from waste and pollution
Symbolic Resonance	The shield is also a symbol of defense and guardianship. Students earning Scudo are not just collecting points — they are building a shield around their campus, their community, and their planet

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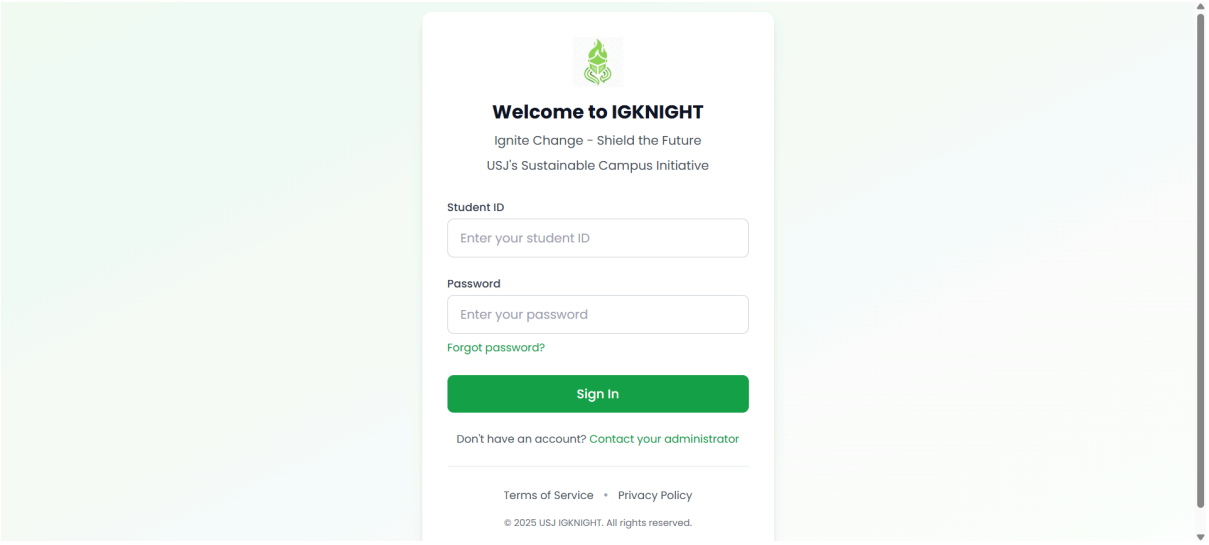
Why This Matters

IGKNIGHT is not a recycling app with a theme painted on top. The Ignatian identity is structural — it shapes how features are designed, how users are motivated, and what outcomes are prioritized.
A standard recycling app says: "Recycle and get points."
IGKNIGHT says: "You are a knight. Your quest is to protect this world. Every bottle you recycle is a shield raised against destruction. Every Scudo you pledge builds something greater than yourself. Rise through the ranks. Earn your crests. See what we accomplish together — for others."
The difference is not cosmetic. It is the difference between a transaction and a transformation.
And that is exactly what happened to Ignatius in 1521 — a single moment that turned a soldier into a saint, a knight into a founder, a wounded man into someone who set the world on fire.
IGKNIGHT invites every USJ student to find their own cannonball moment — and to let it ignite something extraordinary.

"Go forth and set the world on fire."
— St. Ignatius of Loyola

WEB APP DESIGN:


Welcome to IGKNIGHT
Ignite Change - Shield the Future
USJ's Sustainable Campus Initiative
Student ID
Enter your student ID
Password
Enter your password
Forgot password?
Sign In
Don't have an account? Contact your administrator
Terms of Service • Privacy Policy
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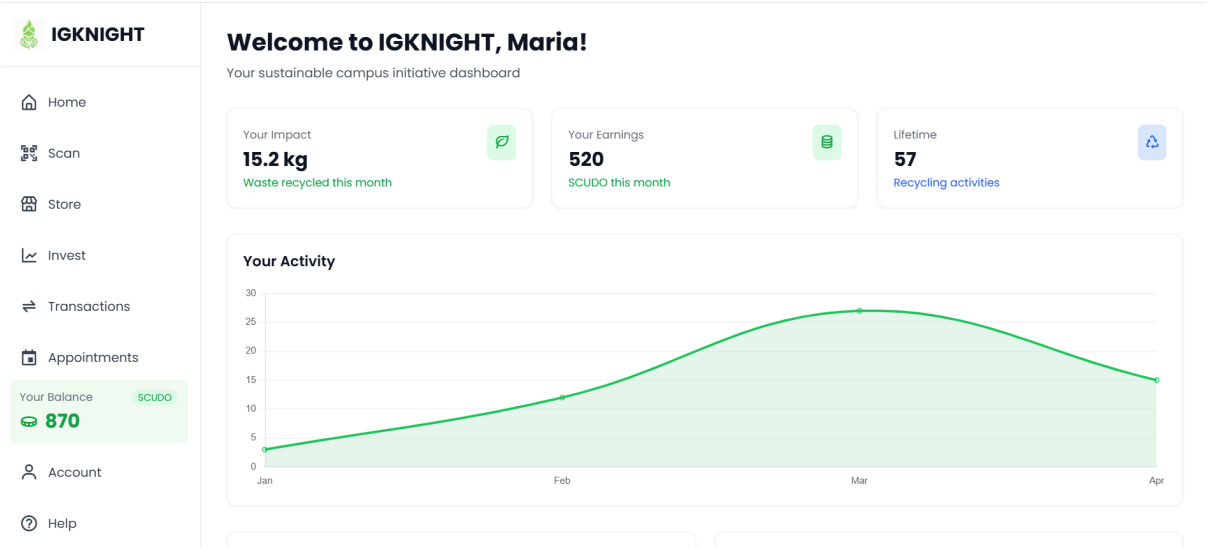
IGKNIGHT
Home
Scan
Store
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Transactions
Appointments
Your Balance
SCUDO
870
Account
Help
Welcome to IGKNIGHT, Maria!
Your sustainable campus initiative dashboard
Your Impact
15.2 kg
Waste recycled this month
Your Earnings
520
SCUDO this month
Lifetime
57
Recycling activities
Your Activity
30
25
20
15
10
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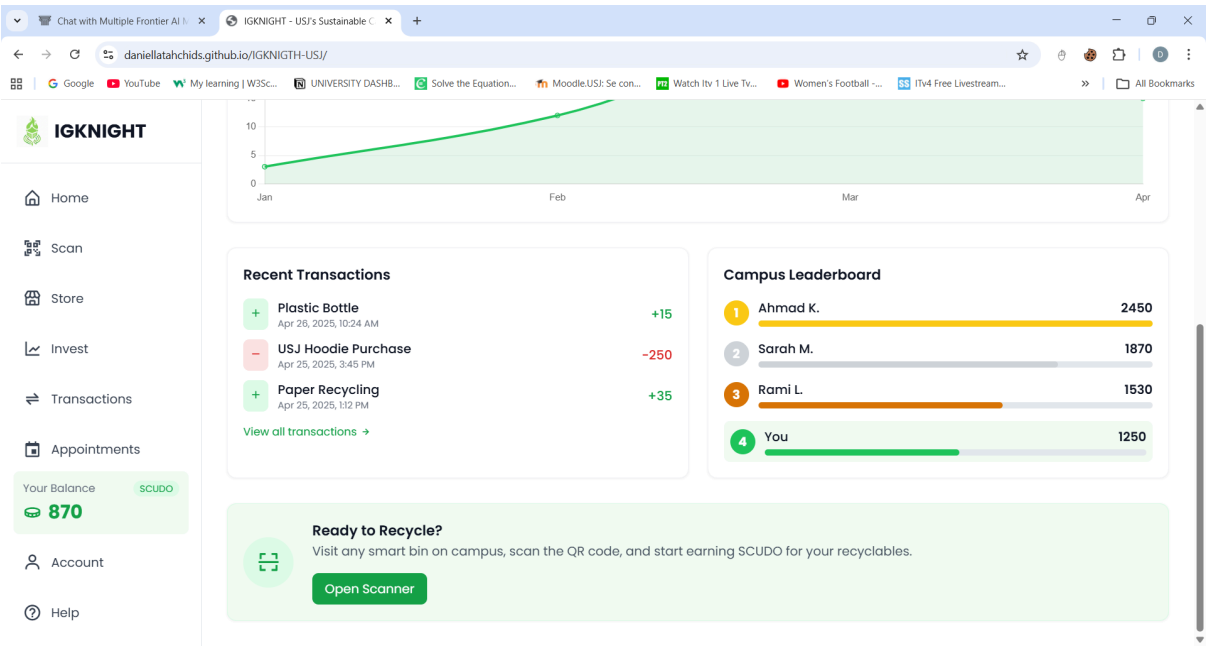
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All Bookmarks
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Appointments
Your Balance
SCUDO
870
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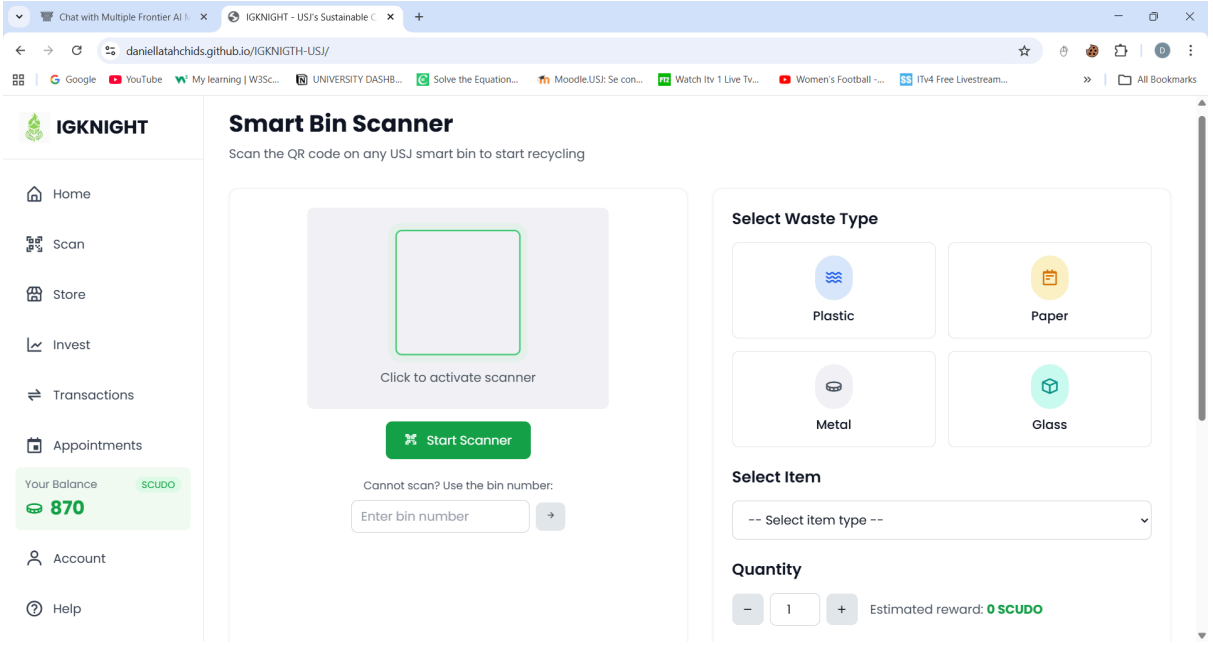
Help
10
5
0
Jan
Feb
Mar
Apr
Recent Transactions
Plastic Bottle
Apr 26, 2025, 10:24 AM
+15
USJ Hoodie Purchase
Apr 25, 2025, 3:45 PM
-250
Paper Recycling
Apr 25, 2025, 1:12 PM
+35
View all transactions →
Campus Leaderboard
1 Ahmad K. 2450
2 Sarah M. 1870
3 Rami L. 1530
4 You 1250
Ready to Recycle?
Visit any smart bin on campus, scan the QR code, and start earning SCUDO for your recyclables.
Open Scanner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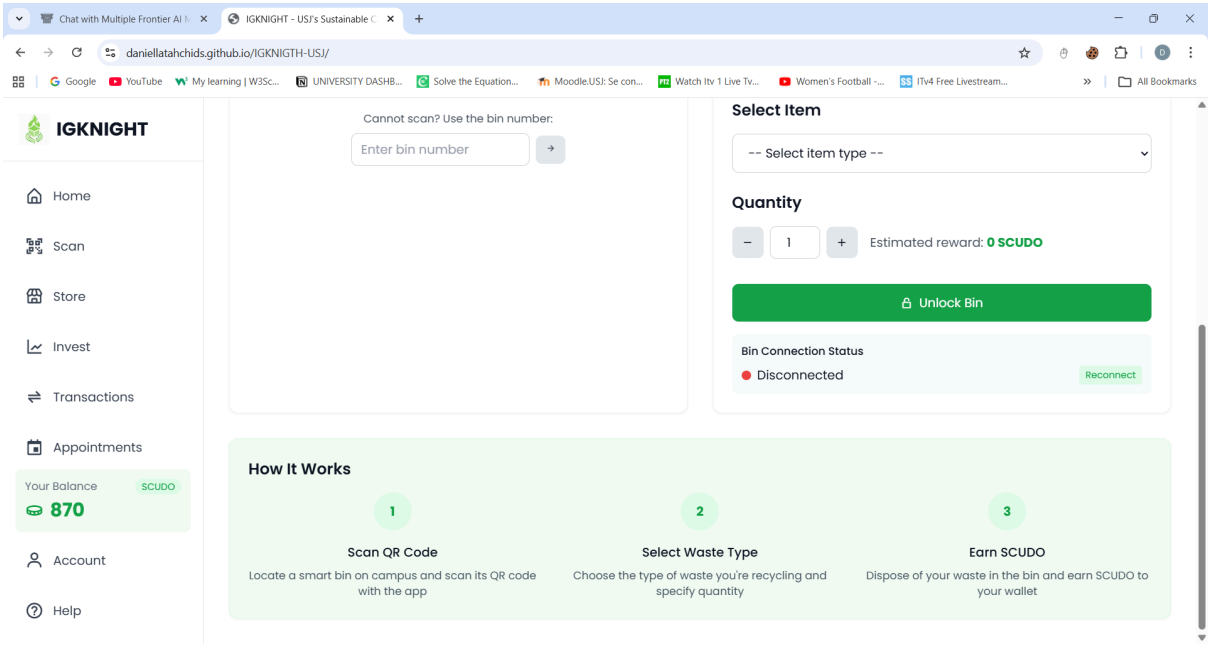
Help
Smart Bin Scanner
Scan the QR code on any USJ smart bin to start recycling
Click to activate scanner
Start Scanner
Cannot scan? Use the bin number:
Enter bin number
Select Waste Type
Plastic
Paper
Metal
Glass
Select Item
-- Select item type --
Quantity
1
Estimated reward: 0 SCUDO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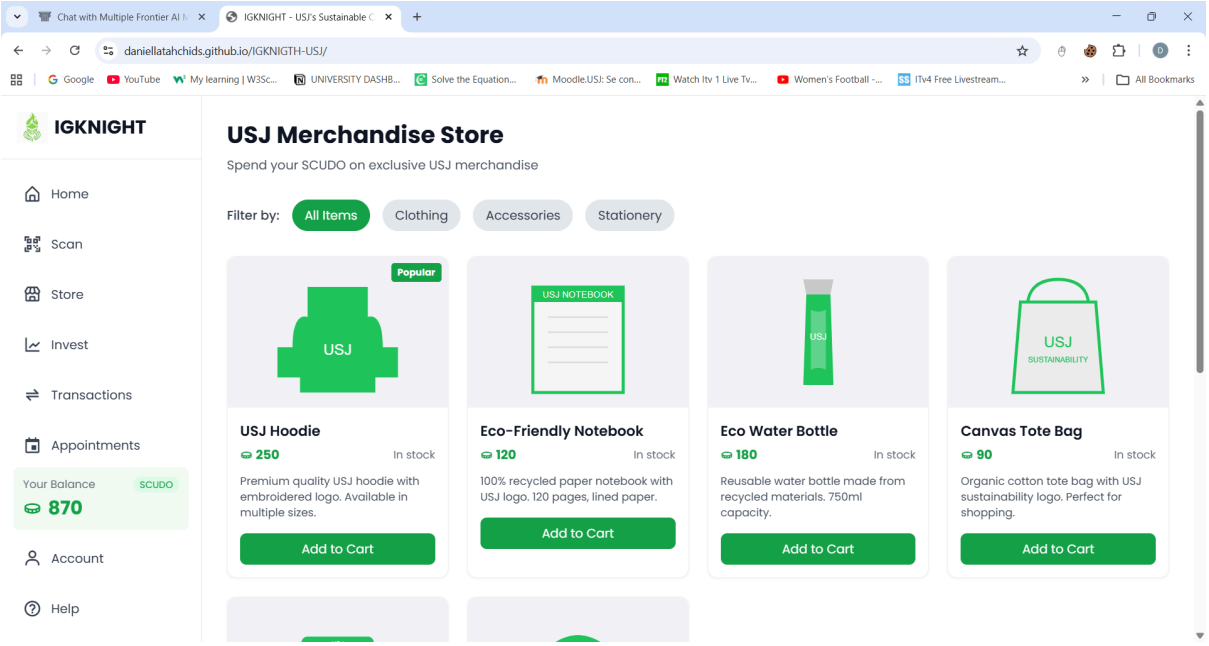
Help
Cannot scan? Use the bin number:
Enter bin number
Select Item
-- Select item type --
Quantity
1
Estimated reward: 0 SCUDO
Unlock Bin
Bin Connection Status
Disconnected
Reconnect
How It Works
1
Scan QR Code
Locate a smart bin on campus and scan its QR code with the app
2
Select Waste Type
Choose the type of waste you're recycling and specify quantity
3
Earn SCUDO
Dispose of your waste in the bin and earn SCUDO to your wallet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SCUDO
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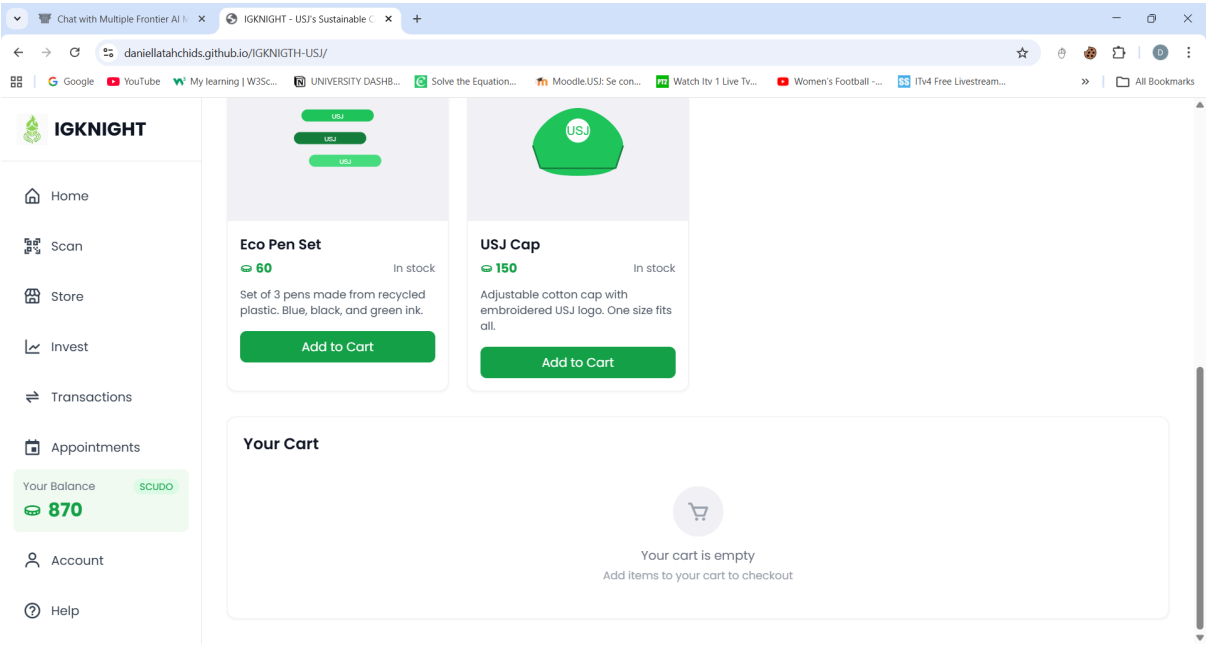
Account
Help
USJ Merchandise Store
Spend your SCUDO on exclusive USJ merchandise
Filter by:
All Items
Clothing
Accessories
Stationery
Popular
USJ
USJ Hoodie
250
In stock
Premium quality USJ hoodie with embroidered logo. Available in multiple sizes.
Add to Cart
USJ NOTEBOOK
Eco-Friendly Notebook
120
In stock
100% recycled paper notebook with USJ logo. 120 pages, lined paper.
Add to Cart
Eco Water Bottle
180
In stock
Reusable water bottle made from recycled materials. 750ml capacity.
Add to Cart
USJ
SUSTAINABILITY
Canvas Tote Bag
90
In stock
Organic cotton tote bag with USJ sustainability logo. Perfect for shopping.
Add to Cart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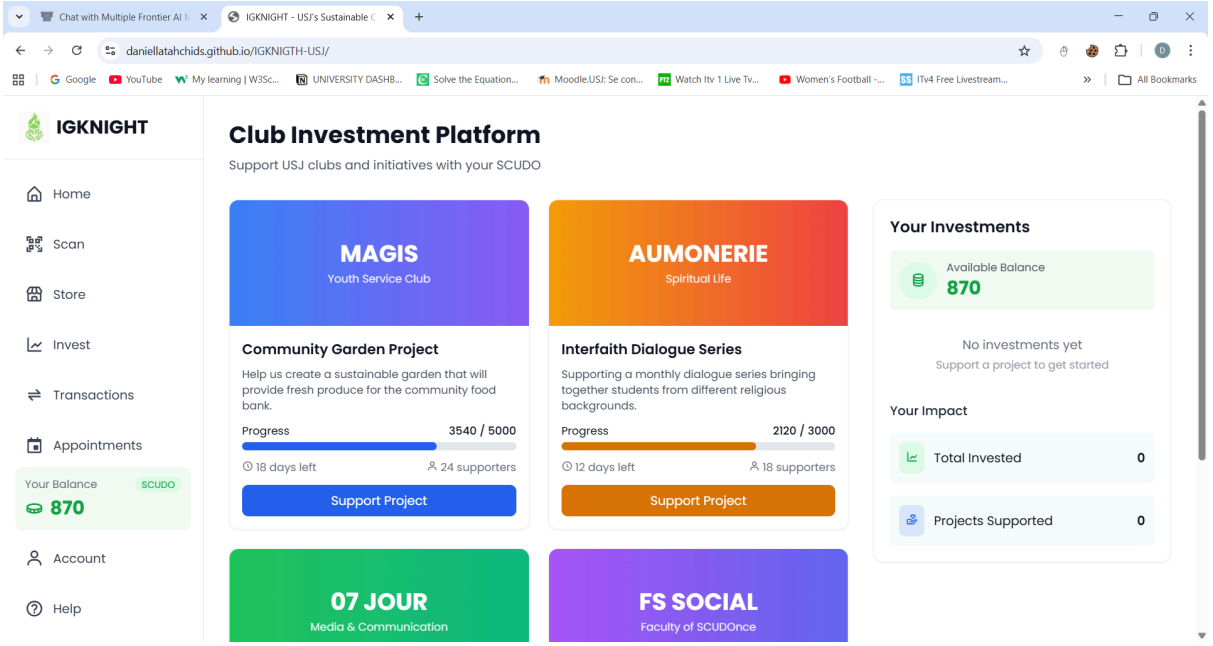
Help
Eco Pen Set
60
In stock
Set of 3 pens made from recycled plastic. Blue, black, and green ink.
Add to Cart
USJ
USJ Cap
150
In stock
Adjustable cotton cap with embroidered USJ logo. One size fits all.
Add to Cart
Your Cart
Your cart is empty
Add items to your cart to checkout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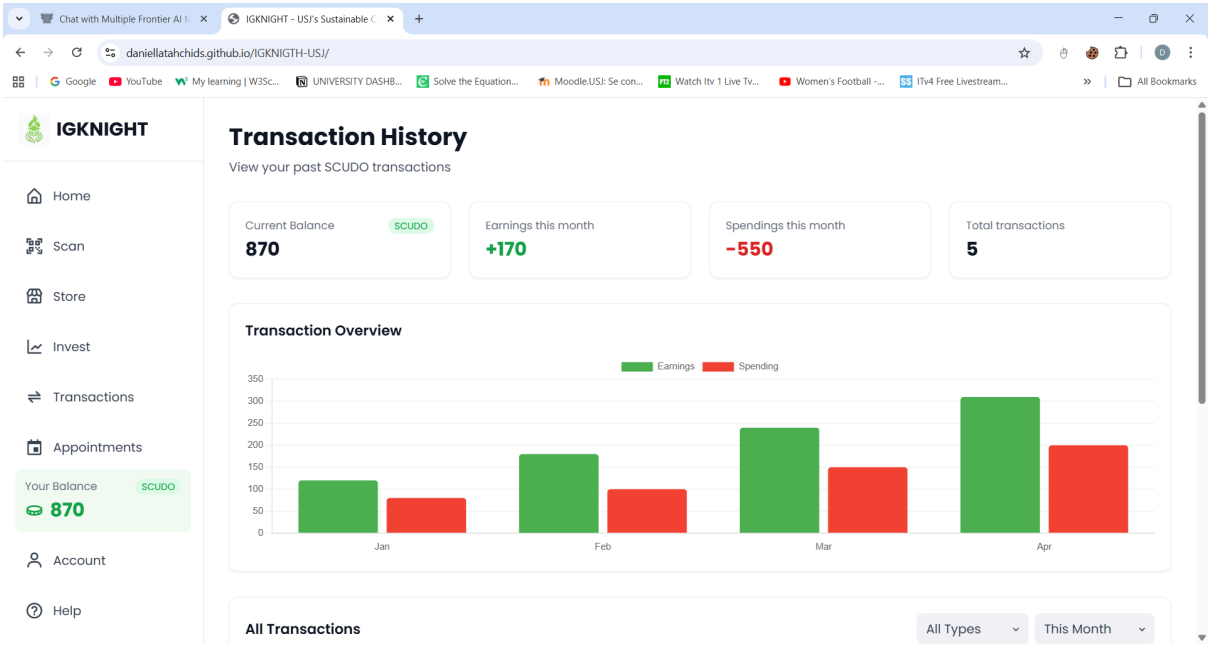
Account
Help
Club Investment Platform
Support USJ clubs and initiatives with your SCUDO
MAGIS
Youth Service Club
Community Garden Project
Help us create a sustainable garden that will provide fresh produce for the community food bank.
Progress
3540 / 5000
18 days left
24 supporters
Support Project
AUMONERIE
Spiritual Life
Interfaith Dialogue Series
Supporting a monthly dialogue series bringing together students from different religious backgrounds.
Progress
2120 / 3000
12 days left
18 supporters
Support Project
07 JOUR
Media & Communication
FS SOCIAL
Faculty of SCUDOnce
Your Investments
Available Balance
870
No investments yet
Support a project to get started
Your Impact
Total Invested
0
Projects Supported
0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SCUDO
870
Account
Help
Transaction History
View your past SCUDO transactions
Current Balance
SCUDO
870
Earnings this month
+170
Spendings this month
-550
Total transactions
5
Transaction Overview
Earnings
Spending
350
300
250
200
150
100
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Jan
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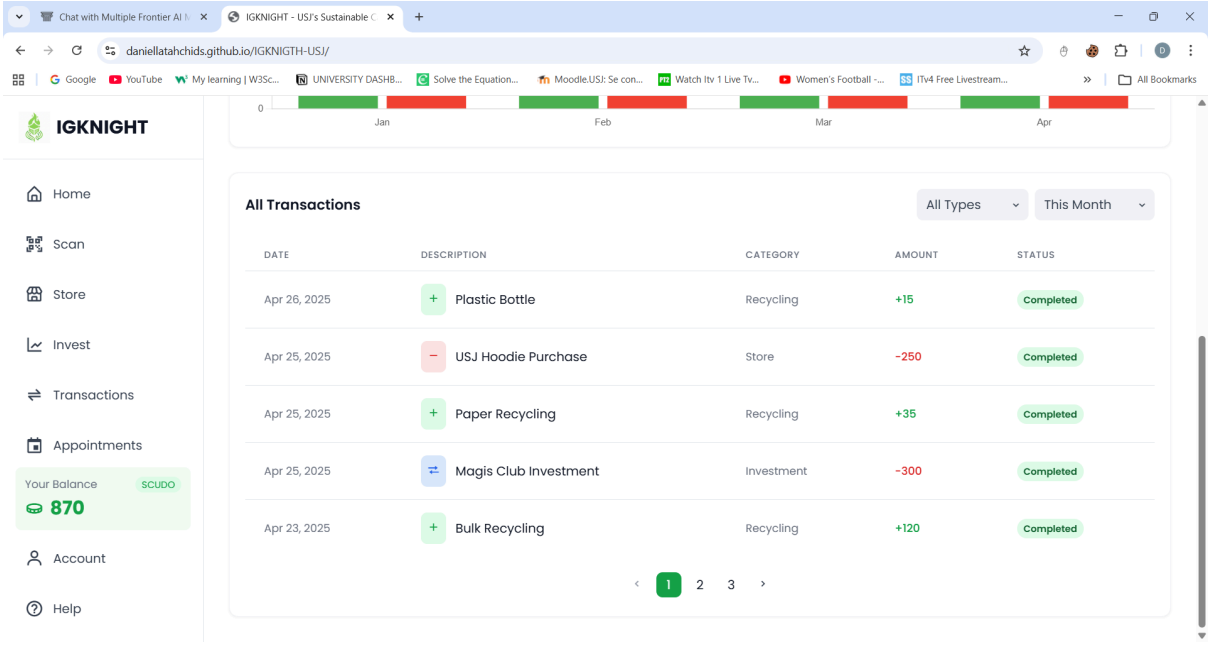
All Transactions
All Types
This Month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Jan
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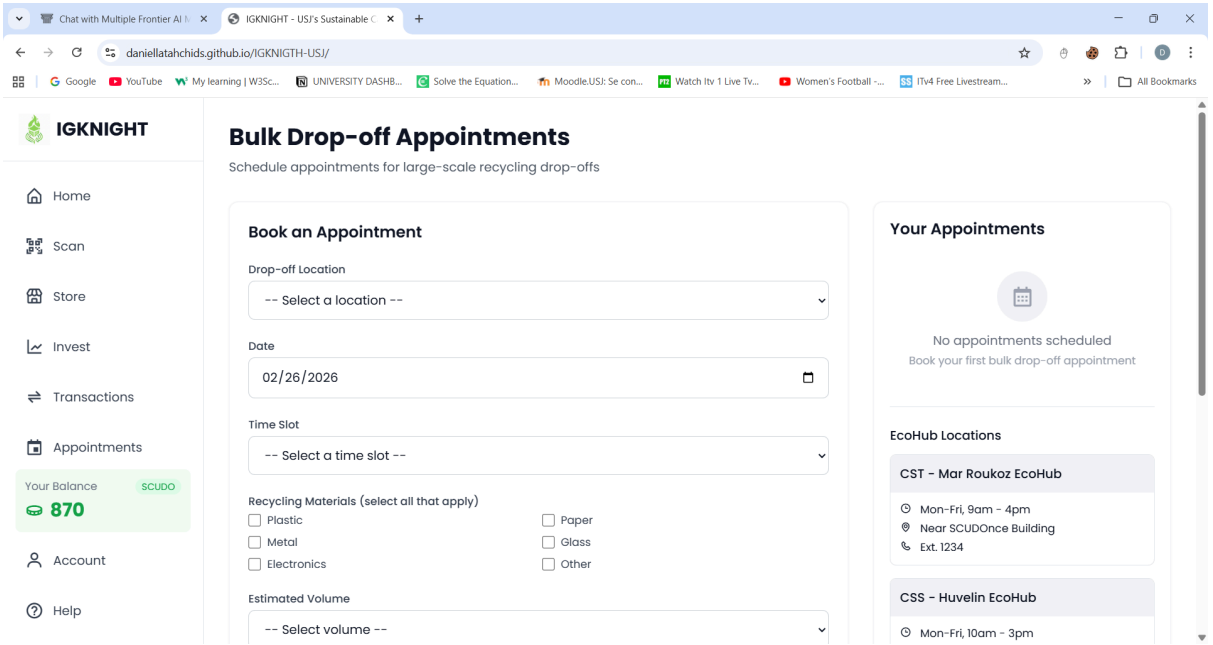
All Transactions
All Types
This Month
DATE
DESCRIPTION
CATEGORY
AMOUNT
STATUS
Apr 26, 2025
Plastic Bottle
Recycling
+15
Completed
Apr 25, 2025
USJ Hoodie Purchase
Store
-250
Completed
Apr 25, 2025
Paper Recycling
Recycling
+35
Completed
Apr 25, 2025
Magis Club Investment
Investment
-300
Completed
Apr 23, 2025
Bulk Recycling
Recycling
+120
Completed
1
2
3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Your Balance
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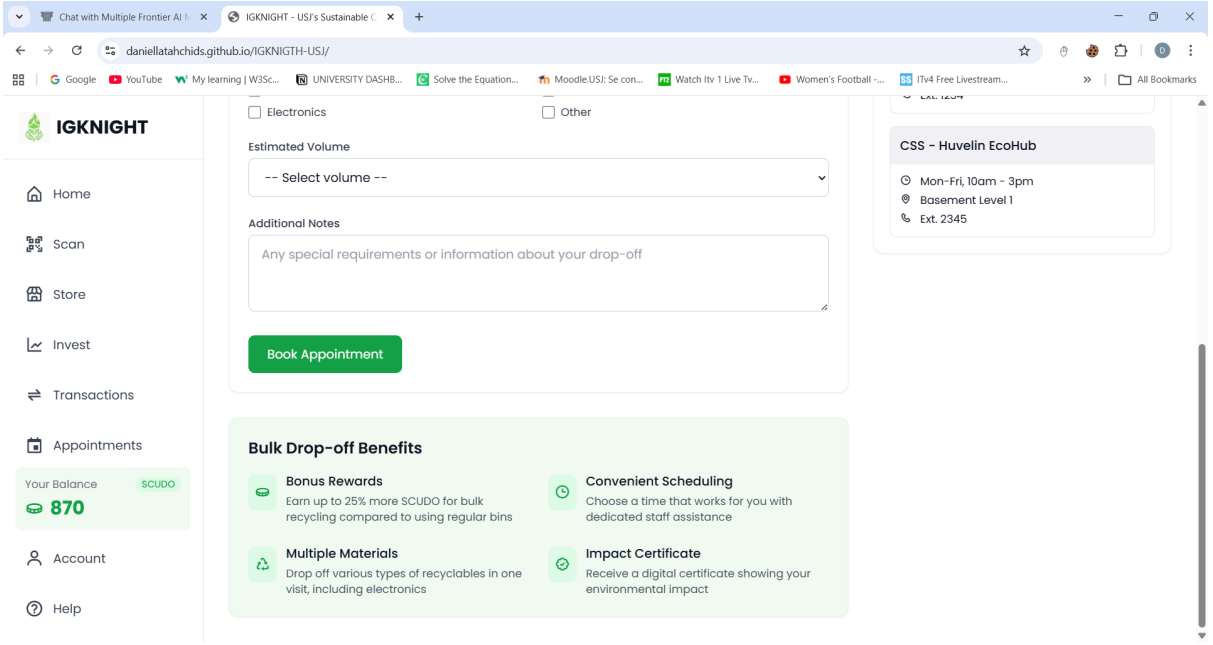
Account
Help
Bulk Drop-off Appointments
Schedule appointments for large-scale recycling drop-offs
Book an Appointment
Drop-off Location
-- Select a location --
Date
02/26/2026
Time Slot
-- Select a time slot --
Recycling Materials (select all that apply)
Plastic
Paper
Metal
Glass
Electronics
Other
Estimated Volume
-- Select volume --
Your Appointments
No appointments scheduled
Book your first bulk drop-off appointment
EcoHub Locations
CST - Mar Roukoz EcoHub
Mon-Fri, 9am - 4pm
Near SCUDOnce Building
Ext. 1234
CSS - Huvelin EcoHub
Mon-Fri, 10am - 3pm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Your Balance
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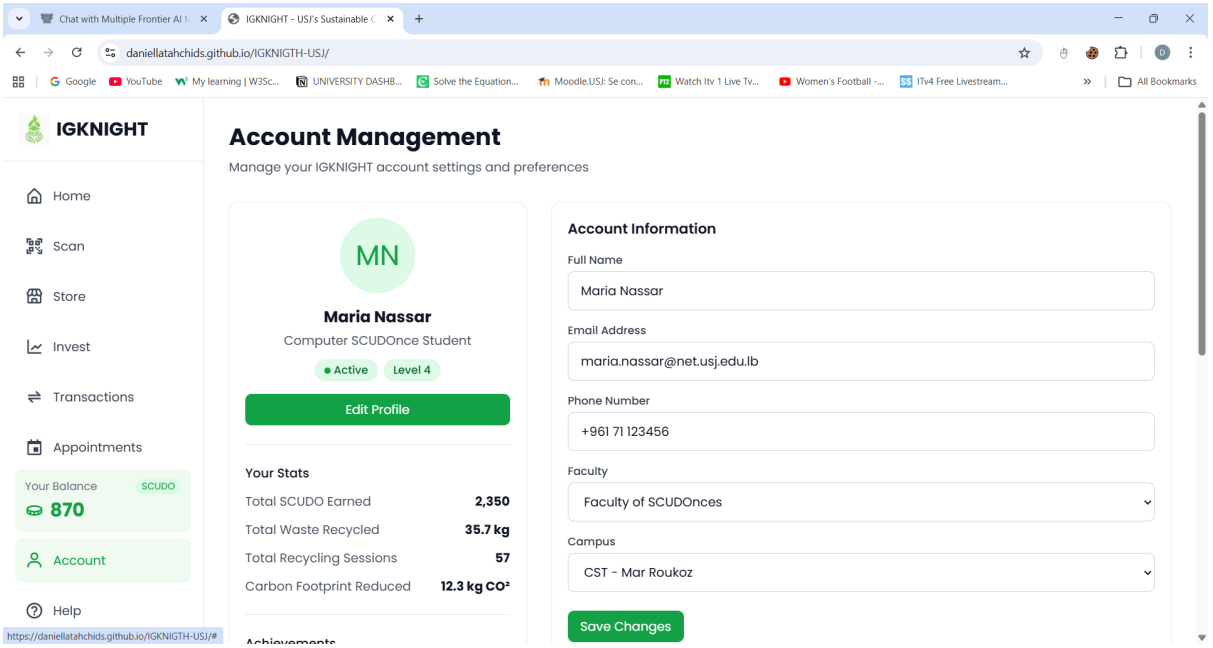
Account
Help
Electronics
Other
Estimated Volume
-- Select volume --
Additional Notes
Any special requirements or information about your drop-off
Book Appointment
CSS - Huvelin EcoHub
Mon-Fri, 10am - 3pm
Basement Level 1
Ext. 2345
Bulk Drop-off Benefits
Bonus Rewards
Earn up to 25% more SCUDO for bulk recycling compared to using regular bins
Convenient Scheduling
Choose a time that works for you with dedicated staff assistance
Multiple Materials
Drop off various types of recyclables in one visit, including electronics
Impact Certificate
Receive a digital certificate showing your environmental impact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Your Balance
SCUDO
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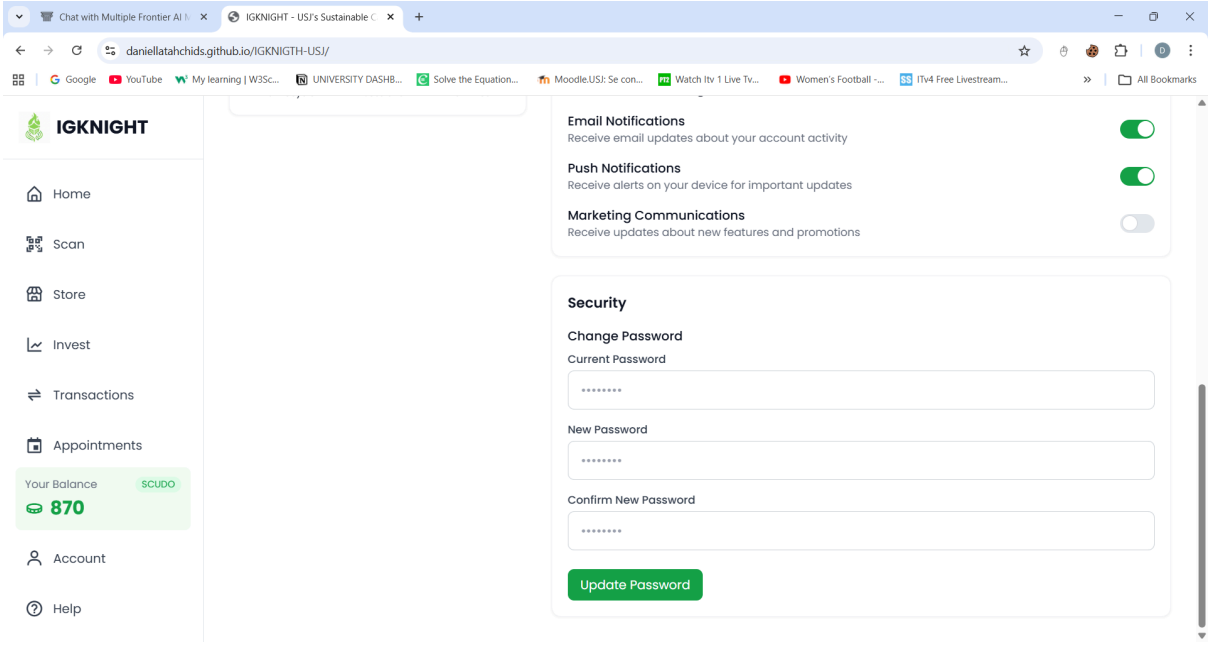
Account
Help
https://daniellatahchids.github.io/IGKNIGTH-USJ/#
Account Management
Manage your IGKNIGHT account settings and preferences
MN
Maria Nassar
Computer SCUDOnce Student
Active
Level 4
Edit Profile
Your Stats
Total SCUDO Earned 2,350
Total Waste Recycled 35.7 kg
Total Recycling Sessions 57
Carbon Footprint Reduced 12.3 kg CO²
Account Information
Full Name
Maria Nassar
Email Address
maria.nassar@net.usj.edu.lb
Phone Number
+961 71 123456
Faculty
Faculty of SCUDOnces
Campus
CST - Mar Roukoz
Save Changes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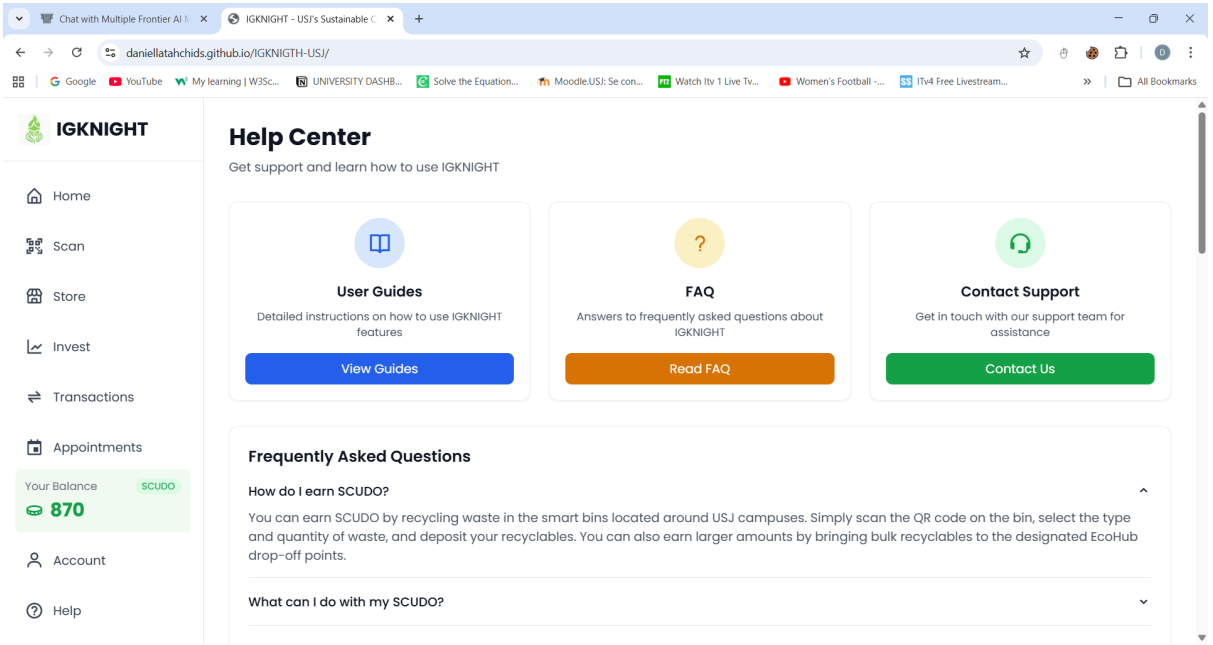
Account
Help
Email Notifications
Receive email updates about your account activity
Push Notifications
Receive alerts on your device for important updates
Marketing Communications
Receive updates about new features and promotions
Security
Change Password
Current Password
New Password
Confirm New Password
Update Password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Account
Help
Help Center
Get support and learn how to use IGKNIGHT
User Guides
Detailed instructions on how to use IGKNIGHT features
View Guides
FAQ
Answers to frequently asked questions about IGKNIGHT
Read FAQ
Contact Support
Get in touch with our support team for assistance
Contact Us
Frequently Asked Questions
How do I earn SCUDO?
You can earn SCUDO by recycling waste in the smart bins located around USJ campuses. Simply scan the QR code on the bin, select the type and quantity of waste, and deposit your recyclables. You can also earn larger amounts by bringing bulk recyclables to the designated EcoHub drop-off points.
What can I do with my SCUDO?

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IGKNIGHT

- Home
- Scan
- Store
- Invest
- Transactions
- Appointments

Your Balance SCUDO
870

- Account
- Help

Volunteer Application

Join our team of dedicated volunteers and help make USJ a more sustainable campus!

First Name

Last Name

Email Address

Phone Number

Faculty/Department

-- Select your faculty --

Areas of Interest (select all that apply)

- [ ] Technology & App Development
- [ ] Events & Campaigns
- [ ] Education & Awareness
- [ ] Waste Management
- [ ] Marketing & Communications
- [ ] Administration & Logistics

Availability

-- Select your availability --

Why do you want to volunteer with IGKNIGHT?

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Skills & Experience (optional)

☐ I agree to the volunteer terms and confirm my availability for the selected hours

Submit Application

Contact Support

Full Name

Email Address

Subject

Message

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